

YouTubeTurboStudio

Architecture Addendum V1 - Originality / Monetization Guard + Final Long / Shorts / Linked-Shorts Model

Purpose: Apply this document to the architecture that is already implemented. This is a delta/addendum only. Antigravity must preserve working code and make the smallest maintainable changes required.

1. Mandatory Final Content Modes

The generation UI for every channel must expose exactly three primary content modes:

Mode	Behavior
Long Video	Generate a standalone long-form video. After it is rendered and/or published, store it as an eligible parent for linking.
Shorts	Generate a fully standalone Short. It must not require, reference, or link to any Long video.
Shorts - Link Long Video	Generate purpose-built Shorts associated with a selected Long video from the active channel. This mode is disabled if no eligible Long video exists.

Required UI behavior

```
ACTIVE CHANNEL: [selected channel]

CONTENT TYPE:
( ) Long Video
( ) Shorts
( ) Shorts - Link Long Video

If no eligible Long video exists:
[locked] Shorts - Link Long Video
Message: Generate, import, or publish an eligible Long video first.

If eligible Long videos exist:
Parent Long Video: [dropdown filtered to active channel only]
Number of Shorts: [1..N]
Generation Mode: [Purpose-built | Extract | Hybrid]

Options:
[x] Generate unique hook for every Short
[x] Preserve standalone value
[x] Add parent Long-video URL to description when URL exists
[x] Add contextual "Watch Full Video" CTA
[x] Keep in same content family
[x] Prevent duplicate Short concepts
```

Cross-channel linking is prohibited. A Kids Short cannot link to an Elders or The AI Brief It Long video, and vice versa.

2. Long Video to Linked Shorts Relationship

Introduce a shared Content Family model above both rendering engines. The relationship logic is shared across The AI Brief It, Kids, Elders, and future channels.

```
LONG VIDEO (parent)
|
+-- Linked Short 1
+-- Linked Short 2
+-- Linked Short 3

Required identifiers:
channel_id
content_id
content_family_id
content_type = LONG | SHORT
relationship_type = STANDALONE | PARENT | DERIVED
parent_content_id
youtube_video_id
```

youtube_url

Linked Shorts should normally be purpose-built from the strongest subtopics, moments, lessons, or story beats in the parent Long video. Do not merely crop random sections by default.

If the parent is already published, the Short description must use the real stored YouTube URL. If the parent has not yet been published, the system must not fabricate a URL. Either publish/schedule the parent first or defer URL injection until the real URL is available.

3. Originality Engine - Hard Publishing Gate

Successful generation or rendering does not make a video publishable. Every video must be evaluated against historical content from the same channel.

Create a separate Originality Score from 0-100. Initial default threshold: 85, configurable per channel.

Check	Purpose
Script similarity	Detect near-duplicate narration, sentence structures, and repeated explanations.
Concept similarity	Detect repeated ideas where only nouns/numbers/names changed.
Hook repetition	Prevent the same hook structure from dominating the channel.
Story structure	Detect repeated plot beats, especially for Kids and Elders.
Visual / scene reuse	Detect excessive reuse of identical visual sequences.
Animation action reuse	Detect repeated action chains such as walk-talk-point-jump across many episodes.
Title / metadata similarity	Prevent minor title variations from becoming mass-produced series.
Meaningful value-add	Confirm the video provides distinct educational, informational, or entertainment value.
Asset provenance	Track where media, music, characters, and generated assets came from.

Do not treat recurring branded characters as inherently repetitive. Same character + genuinely new story/problem/learning objective/environment/interactions is acceptable. Same template with only superficial substitutions should be penalized.

4. Production Quality and Compliance Gates

Keep Production Quality separate from Originality. Initial Production Quality target: 90/100, configurable.

READY FOR REVIEW only when:

```
Production Quality >= configured threshold
AND Originality >= configured threshold
AND Technical QC = PASS
AND Shared Content Compliance = PASS
AND Channel-specific validation = PASS
```

Kids additionally requires:

```
Kids Safety = PASS
Educational Accuracy = PASS
Made-for-Kids configuration = valid
```

Critical rule: publishing frequency is a maximum target, never a quota. Never lower quality, originality, safety, or compliance thresholds to meet a daily schedule.

5. Targeted Regeneration

When a gate fails, regenerate only the failing component where practical:

```
Hook similarity failure -> regenerate hook
Script originality failure -> rewrite script
Story similarity failure -> regenerate plot/story structure
Visual repetition failure -> regenerate storyboard / visual plan
Animation repetition failure -> change actions, camera, staging, environment
Caption failure -> regenerate captions only
Voice/audio failure -> regenerate audio only
Technical render failure -> rerender
```

Kids safety failure -> block and regenerate affected content
Compliance uncertainty -> manual review
Asset provenance failure -> replace unverified asset

6. Channel-Specific Behavior

The AI Brief It

Continue using the Media Video Engine. Improve originality through distinct science explanations, relevant Pexels video-first selection, Pexels images as fallback, diagrams, motion graphics, kinetic captions, better audio, and varied visual sequencing. Avoid repeated image-zoom-caption templates.

Kids

Use the shared Animation Engine plus mandatory Kids safeguards. Validate age appropriateness, educational correctness, exact counts/letters/colors where applicable, safe language/visuals, original or properly licensed characters/assets, and correct Made-for-Kids handling. Do not rely on comments for engagement.

Elders

Use the same Animation Engine code as Kids, but adult assets, voices, prompts, pacing, environments, and story structures. Do not inherit Kids-specific safety/educational gates unless independently applicable.

7. Content History Required for Real Originality Checks

Persist enough information to compare new content against prior content from the same channel:

```
channel_id
content_id
content_family_id
content_type
relationship_type
parent_content_id

topic
concept
hook
hook_type
script
script fingerprint / embedding / similarity representation
story structure

visual plan
scene signatures
animation actions
assets used
music used
voice profile

title
description

production_quality_score
originality_score
technical_qc_result
compliance_result
channel_validator_result

youtube_video_id
youtube_url
publish_timestamp
analytics snapshots
```

Do not rely only on an LLM asking itself whether content is original. Use deterministic similarity/fingerprint checks where practical, with LLM review as an additional layer.

8. Publishing and Metadata Rules for Linked Shorts

When publishing a linked Short:

1. Resolve active channel.
2. Resolve parent Long video from same channel.
3. Verify parent relationship.
4. If parent has a real YouTube URL:
 - add contextual full-video link to Short description.
5. If parent URL does not yet exist:
 - never fabricate one.
 - publish/schedule parent first or defer link injection.
6. Generate Short-specific title and description.
7. Preserve standalone viewer value.
8. Add contextual CTA to full video where appropriate.
9. Use YouTube related-video association if supported by the available API/workflow.
10. Do not depend on pinned comments for Kids content.

Standalone Shorts must not accidentally inherit a parent URL or content-family CTA.

9. Dashboard / Internal Readiness

Add an internal readiness panel. It must never claim guaranteed YouTube monetization.

```
PRODUCTION QUALITY 93/100 PASS
ORIGINALITY 89/100 PASS
TECHNICAL QC PASS
CONTENT COMPLIANCE PASS
CHANNEL VALIDATION PASS
KIDS SAFETY N/A or PASS
ASSET PROVENANCE PASS
```

```
INTERNAL STATUS:
READY FOR REVIEW
```

10. Acceptance Tests

1. Standalone Short can be generated with no parent relationship and no parent URL.
2. Long Video can be generated and stored as an eligible parent.
3. Shorts - Link Long Video is disabled when no eligible Long exists.
4. Linked mode becomes enabled when an eligible Long exists.
5. Parent dropdown contains only Long videos from the active channel.
6. Cross-channel parent selection is rejected server-side even if UI is bypassed.
7. Linked Short stores parent_content_id and content_family_id.
8. Published parent URL is automatically inserted into linked Short description.
9. No fake URL is created for an unpublished parent.
10. Standalone Short never receives a parent URL.
11. Originality failure blocks publishing.
12. Production Quality failure blocks publishing.
13. Technical QC failure blocks publishing.
14. Kids safety/educational failure blocks Kids publishing.
15. Upload target never overrides failed gates.
16. Targeted regeneration changes only the failed stage where feasible.
17. Recurring characters are allowed while near-duplicate stories/templates are detected.
18. All settings remain isolated by channel.

11. Antigravity Implementation Instruction

The existing architecture is already generated. Do not rebuild the application. Read the existing AGENTS.md and MASTER_ARCHITECTURE.md first, then apply this addendum as a newer requirement where conflicts exist. Inspect current code and implement only the minimum required changes. Preserve working channel management, credentials, engines, UI, rendering, and YouTube integration.

Implementation order:

1. Inspect current content models and UI.
2. Add/extend Content Family + parent relationships.
3. Implement the three final content modes.
4. Add Originality Engine and content history.
5. Add separate Production Quality / Compliance gates.
6. Add channel-specific validation hooks.
7. Add targeted regeneration.
8. Update publisher for parent URL injection.
9. Add readiness UI.
10. Add tests and run them.

After implementation report:

- files created
- files modified
- schema/migration changes
- new dependencies
- settings added
- tests executed and results
- API/OAuth changes, if any
- manual steps required
- anything incomplete

Do not publish real YouTube videos during automated tests.

Do not expose or commit credentials.

Do not push directly to main unless explicitly instructed.